

CSCI 160 Homework 3

1. Obtain the truth table of the function $F(x, y, z) = x'z + yz + xy'$. Using the minterm rows, fill in the three-variable K-map and simplify F to a minimal sum-of-products expression.
2. Simplify the Boolean function $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 11, 14)$ using a four-variable K-map. Identify all prime implicants, determine which are essential, and write the minimal sum-of-products expression.
3. Simplify the Boolean function

$$F(A, B, C, D) = \prod M(1, 3, 5, 7, 9, 11, 13, 15)$$

using a four-variable K-map. Express the minimized result as a product-of-sums expression.

4. Simplify the following Boolean function into (a) sum-of-products form and (b) product-of-sums form:

$$G(w, x, y, z) = \sum m(0, 1, 4, 5, 12, 13)$$

5. Simplify the Boolean function

$$F(w, x, y, z) = \sum m(3, 4, 5, 6, 7)$$

which has the don't-care conditions $d(w, x, y, z) = (10, 11, 12, 13, 14, 15)$, to a minimal sum-of-products expression using a four-variable K-map.